ParkinSense: A Continuous Multi-Modal Wearable Platform for Motion, Cardiac, and Activity Digital Biomarkers in Parkinson's Disease Monitoring

Pranshu Kumar, Lakshya Lalwani, Akshay Sable, Nalin Khare, Asim Tewari
Department of Mechanical Engineering, Indian Institute of Technology Bombay
Mumbai 400076, India

Abstract*—Parkinson's disease (PD) is conventionally assessed through intermittent clinical visits using rating scales such as the MDS-UPDRS, a sampling regime that misses the fluctuating, activity-dependent nature of motor symptoms and cannot capture medication-cycle variability between appointments. Wearable continuous monitoring has been proposed as a complementary source of objective, high-frequency data, but nearly every reported system – including the recently published ELENA project, which represents the current state of the art in clinically-validated single-modality tremor wearables – isolates a single sensing channel, most often inertial tremor quantification performed offline, without jointly characterizing the cardiovascular and activity context in which tremor occurs. This paper presents ParkinSense, an open-source wrist-worn research platform that departs from this single-modality convention along two independent axes – hardware and algorithms – and runs three synchronized, on-device processing pipelines from a single inertial-plus-optical sensor stream: (i) a motion pipeline performing gravity removal, mains-notch filtering, Parkinsonian-band (4.0–6.5 Hz) Butterworth band-pass filtering, multi-axis joint feature extraction, motion-context classification, and hysteresis-gated tremor state estimation; (ii) a photoplethysmography (PPG) pipeline extracting heart rate, RR intervals, heart-rate variability (RMSSD, SDNN, mean RR, pNN50), and a research-grade SpO_2 estimate via a ratio-of-ratios signal-lock state machine; and (iii) a step-tracking pipeline that derives cadence, distance, walking state, and bout-qualified active minutes from an adaptive-threshold accelerometer detector explicitly gated by the motion pipeline's activity classifier to reject tremor and handling artifacts. Where ELENA reports device-level frequency accuracy from a single MPU6050 axis-independent FFT computed offline in MATLAB after the session ends, ParkinSense computes tremor state within a rolling 4-second window on-device, in real time, from a joint five-feature score across all three gyroscope axes simultaneously, and couples that motion-state estimate into two additional physiological channels that ELENA's architecture has no mechanism to represent at all. We describe the complete signal chain, the hardware rationale for the added optical channel and higher-bandwidth radio link, the cross-pipeline gating architecture, and the system's current experimental evaluation, and we present a structured, criterion-by-criterion comparison against ELENA and other prior wearable and assistive systems. We identify formal clinical validation against labeled ground truth as the primary remaining step before any diagnostic claim can be made, and we outline the dataset-collection and machine-learning roadmap intended to close that gap.

***Index Terms*—Parkinson's disease; wearable sensing; digital biomarkers; tremor detection; photoplethysmography; heart rate variability; inertial measurement unit; step counting; continuous monitoring; Bluetooth Low Energy; edge signal processing; sensor fusion**

I. Introduction

Parkinson's disease is a progressive neurodegenerative disorder affecting an estimated ten million people worldwide, and its cardinal motor signs – resting tremor, bradykinesia, and rigidity – fluctuate over the course of a day in ways that a single clinic visit cannot capture [1], [2]. The MDS-UPDRS and related rating scales remain the clinical standard for grading these symptoms [3], [4], but they are administered infrequently, are subject to inter-rater variability, and necessarily reflect a brief, often atypically alert, in-clinic presentation rather than the patient's ordinary daily motor state. This gap between intermittent clinical grading and the true time-course of PD motor fluctuation has motivated a growing body of work on wearable continuous monitoring [5], most of which has concentrated on a single sensing channel: inertial tremor quantification, typically performed with a single low-cost accelerometer or accelerometer-gyroscope module [6], [7], [19].

A narrower but consistent limitation recurs across nearly the entire wearable-tremor literature, and it is worth stating precisely because it is the central design problem this paper addresses. Tremor amplitude and detectability are known to vary with autonomic state and with whether the wrist is at rest or engaged in voluntary motion [27], yet the great majority of wearable tremor systems – from early clinical accelerometer studies [17], [18] through very recent IoT-connected prototypes – report a single motion-derived score in isolation, without a concurrent cardiac channel to contextualize autonomic state or an activity channel to contextualize whether the wearer was resting, walking, or manipulating an object at the time of the reading. This is not a minor omission. A tremor reading taken while the patient is seated at rest carries a different clinical meaning than the same numerical reading taken while the patient is walking, and neither the amplitude threshold nor the confidence one should place in the reading is invariant to that context. Systems that do not carry a concurrent, independently-derived activity label are structurally unable to make that distinction, no matter how accurate their frequency estimation is in isolation.

The most relevant recent point of comparison is the ELENA project [28], [29], a wrist-and-finger-worn IoT device built around a single MPU6050 accelerometer/gyroscope module and an ESP32 microcontroller, which reports offline MATLAB frequency analysis validated against a six-position static rig, an MTS vibration table, and a nine-patient/nine-control clinical pilot with better than 0.5% frequency deviation against an Apple Watch reference. ELENA is therefore an appropriate and demanding benchmark, not a strawman, and any claim that ParkinSense improves on prior work must be argued specifically against it rather than against a generic "single-sensor wearable." We make that argument along two structurally independent axes:

- **Hardware.** ELENA's sensing is limited to a single 6-axis MPU6050 module streamed over Wi-Fi to a cloud server, with tremor frequency and amplitude extracted afterward in MATLAB. ParkinSense adds a second, independent sensing modality – a MAX30102 optical pulse-oximetry sensor – to the same wrist-worn package, and streams both channels over a single low-latency, low-power 104 Hz Bluetooth Low Energy (BLE) link rather than a Wi-Fi/cloud path, enabling on-device analysis with sub-100 ms latency after each processing window closes rather than an after-the-fact desktop computation.
- **Algorithms.** ELENA's frequency estimate is produced by applying a single Butterworth low-pass filter and an FFT to each axis independently offline, with the dominant peak read off the axis with the highest spectral peak. ParkinSense's motion pipeline instead performs gravity removal, mains-notch filtering, and Parkinsonian-band band-pass filtering before scoring all three gyroscope axes *jointly* on in-band frequency, band-ratio strength, cross-axis agreement, axis dominance, and an explicit motion-context penalty, with a separately-maintained confidence estimate and a hysteresis-based temporal state machine that ELENA's snapshot FFT approach has no equivalent of. Critically, that same motion-context classification is reused as a gating signal by two additional pipelines – cardiac and activity – so that a single, validated notion of "is the wrist actually moving voluntarily" is computed once and shared, rather than each channel re-deriving its own ad hoc notion of motion from scratch, or not accounting for it at all.

The contribution of this paper is accordingly not a single novel filter or classifier, but an integrated, reproducible, three-pipeline wearable architecture in which the hardware and the algorithmic design decisions reinforce one another – the added optical channel is only useful because the runtime has a motion-aware confidence-fusion mechanism to interpret it correctly during movement, and the added activity channel is only trustworthy because it is gated by the same tremor/motion classifier rather than re-solving "is the body moving" independently. We describe the system in enough implementation detail (filter orders, window lengths, state-machine transition conditions, confidence heuristics) to be replicated or extended, we position it explicitly against ELENA and other prior systems on a criterion-by-criterion basis, and we give an honest account of what remains to be clinically validated before any diagnostic claim can be made.

The remainder of this paper is organized as follows. Section II reviews prior wearable and assistive work relevant to PD motor and physiological monitoring, with a subsection contrasting ParkinSense's hardware and algorithmic choices against ELENA point by point. Section III details the ParkinSense system architecture and the three processing pipelines at an implementation level. Section IV specifies the hardware and software stack and the BLE protocol, and justifies each hardware deviation from the ELENA design. Section V reports the platform's experimental evaluation to date. Section VI presents a structured comparative analysis against ELENA and other prior systems. Section VII concludes and outlines future work.

II. Literature Review and Gap Analysis

A. Inertial Tremor Quantification

Wearable inertial sensing for PD tremor has been studied extensively. Wrist- or finger-worn accelerometer and gyroscope systems have demonstrated that resting tremor's spectral signature (4–6 Hz, narrower than the broader 4–12 Hz band associated with essential tremor and voluntary motion below roughly 2 Hz) can be reliably isolated with band-pass filtering and spectral feature extraction [6], [7]. Later work applied neural classifiers to distinguish Parkinsonian tremor from essential tremor and from voluntary motion using inertial features [20], [23], [24], and comparative studies have shown that rhythmicity and amplitude persistence across tasks, not frequency alone, best separate pathological from physiological tremor [21], [22]. Systematic reviews of wearable PD sensing report that most inertial systems output a single tremor severity or score per window, computed independently of any concurrent physiological channel [8], [25], [26].

B. Photoplethysmography and Autonomic Monitoring in Movement Disorders

Autonomic dysfunction is a recognized non-motor feature of PD, and heart-rate variability metrics (RMSSD, SDNN, and related time-domain measures, as standardized by the Task Force of the European Society of Cardiology and the North American Society of Pacing and Electrophysiology [10]) have been proposed as a complementary digital biomarker. Wrist PPG is well established for heart-rate and SpO_2 estimation in consumer and research wearables [11], but PPG-derived HRV and SpO_2 have rarely been fused, in an open research platform, with a concurrent tremor channel under a shared motion-aware confidence model. Larger PD monitoring frameworks such as PERFORM have incorporated physiological channels alongside motion sensing at a systems level [17], but the individual low-cost, single-board wearable tremor prototypes that dominate the recent literature – including ELENA – do not.

C. Inertial Step Counting and Gait Context

Accelerometer-based step detection is a mature area, typically built around high-pass filtering of acceleration magnitude followed by adaptive-threshold peak detection [12]. In the PD-wearable literature, however, activity/step channels are usually reported as a separate, unrelated product feature rather than as a signal that is actively gated by a tremor/motion classifier to prevent involuntary oscillation from being miscounted as gait. Real-world sensor studies of PD patients have shown altered activity patterns, more sedentary time, and fewer state transitions relative to controls [15], [16], underscoring that activity context and tremor context are physiologically linked and ought to be measured jointly rather than by unrelated instruments.

D. Active Compensation Devices

A distinct line of work addresses tremor through active mechanical compensation rather than passive monitoring. Deep-learning-driven voluntary-motion prediction has been used to drive real-time tremor-suppressing actuators [20], and controller-oriented assistive devices such as TremoSense couple a multivariate Extended Kalman Filter with an LQR controller to drive a two-axis stabilized utensil, targeting real-time suppression rather than longitudinal biomarker logging. Such systems solve a different problem – momentary corrective actuation – and are complementary to, rather than a substitute for, a monitoring platform such as ParkinSense; a system like ParkinSense could, in principle, supply the motion-context classification that an actuation system needs to distinguish tremor from intended motion.

E. The ELENA Project: A Point-by-Point Comparison

Because ELENA [28] is the closest and most rigorously validated prior system, we compare against it in more depth than a typical related-work paragraph, while keeping the discussion focused on the structural differences that motivate ParkinSense's design. ELENA's hardware is an Adafruit HUZZAH32 (ESP32) with a single MPU6050 6-DoF module mounted on the index finger, a microSD card, an OLED display, and a Wi-Fi uplink to a cloud platform. Tremor analysis is performed entirely offline: recordings are imported into MATLAB, denoised with a Butterworth low-pass and a 1 Hz high-pass filter, and transformed with an FFT, with the axis carrying the highest spectral peak taken as the dominant tremor axis. The system was calibrated on a six-position static rig and an MTS uniaxial vibration table across 0.5–10 Hz, benchmarked against an Apple Watch Series 9 tremor-analysis application to within 0.5% mean frequency deviation, and evaluated on nine PD patients and nine age-matched controls under an IRB-approved protocol (CNBI EC-CNBI-2023-02-176), with dominant frequencies observed in the 3–7 Hz band at rest. This is a strong result, and ELENA's own authors acknowledge voluntary-motion artifact during active tasks as an open limitation, with several patient records containing unresolved entries where the tremor signal was too contaminated by voluntary motion to extract cleanly.

We highlight three structural properties of the ELENA architecture that follow from its single-modality, offline design, each of which motivates a specific ParkinSense design decision. First, ELENA's only sensor is the MPU6050, so there is no channel available to indicate whether a recording window coincided with elevated heart rate, reduced heart-rate variability, or ambulatory movement elsewhere in the body; ParkinSense's added MAX30102 optical sensor and step-tracking pipeline are a direct response, fused into a single motion-aware confidence framework rather than reported as unrelated numbers. Second, ELENA's MATLAB pipeline runs after a session ends, with no on-device live state, confidence score, or temporal filter to prevent a single noisy window from flipping the reported condition; ParkinSense instead computes a composite tremor score every 4 seconds on-device and only confirms a state transition after five positive windows in a rolling ten-window history. Third, ELENA selects the axis with the largest independent spectral peak, discarding the cross-axis agreement that distinguishes a true rotational tremor from a spurious single-axis artifact such as a knock against a table edge; ParkinSense scores all three gyroscope axes jointly, with explicit cross-axis agreement, axis-dominance, and active-state penalty terms.

An honest comparison must also state where ELENA is ahead: it has completed a nine-patient, nine-control clinical pilot under an approved bioethics protocol with a treating neurologist's task protocol, and its device-level frequency accuracy against a commercial reference is a concrete, reproducible number that ParkinSense does not yet have an equivalent of. The two systems are therefore best read as addressing different parts of the same overall validation problem: ELENA demonstrates what a completed single-modality clinical pilot looks like and is the closer near-term model for how ParkinSense's own planned pilot (Section V) should be designed and reported, while ParkinSense's contribution is the three-pipeline, on-device, hardware-plus-algorithm architecture that a future clinical study would need to be run against to show whether the added modalities and joint scoring actually improve on ELENA's single-axis offline baseline in a patient population, rather than only in bench testing.

F. Consolidated Gap Summary

Table I summarizes the limitations that recur across the reviewed systems and the corresponding design response in ParkinSense.

III. Proposed System: ParkinSense Architecture

A. System Overview and Signal Flow

ParkinSense is a wrist-worn, continuous, three-pipeline monitoring platform. A single binary BLE packet stream, decoded once by the runtime, feeds a motion pipeline, a PPG pipeline, and a step-tracking pipeline in parallel every cycle. Fig. 1 illustrates the signal flow from sensor acquisition through digital biomarker fusion to logging and visualization, and Fig. 2 gives the corresponding stage-by-stage view of all three pipelines together with the core algorithmic relations used at each stage.

TABLE I: Consolidated Limitation Analysis of Prior Wearable PD Monitoring Approaches

Approach	Modality	Key Limitation
Inertial tremor wearables [6], [7]	Motion only	No concurrent physiological or activity context
Consumer PPG wearables [11]	Cardiac only	No tremor channel; HRV rarely exposed
Inertial step counters [12]	Activity only	Not gated against tremor/handling artifacts
ELENA [28]	Motion only (single MPU6050, per-axis FFT)	Offline MATLAB analysis, no live state or confidence estimate; single-axis peak selection; no activity/cardiac context
PERFORM [17]	Multi-sensor, system-level	Clinical framework rather than a single reproducible open-source wearable
Active compensation (e.g. TremoSense, [20])	Motion + control	Targets real-time actuation, not longitudinal biomarker logging
ParkinSense	Motion + PPG + activity	Three pipelines from one synchronized stream, real-time on-device, joint multi-axis scoring, activity/cardiac explicitly gated by shared motion state; clinical validation still pending

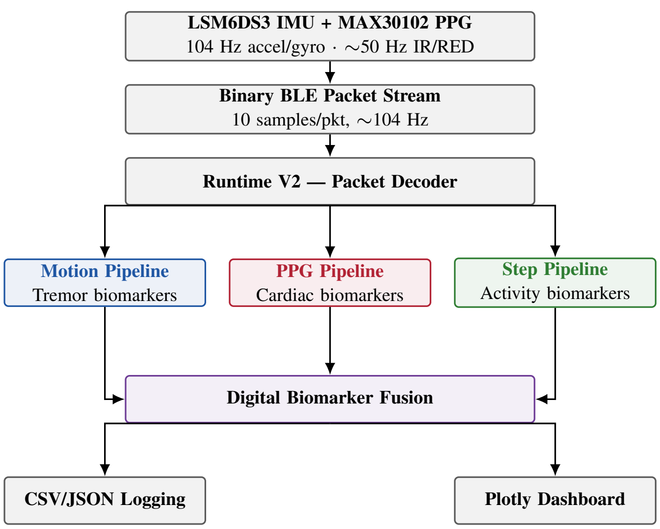


Fig. 1: ParkinSense signal flow: three color-coded pipelines consume a single decoded sensor stream and publish to a shared logging and visualization layer.

Each pipeline is single-responsibility and independently testable, consistent with the platform's design goal of supporting replay-based validation and future machine-learning integration without coupling the sensing, analytics, and presentation layers together. This is a deliberate architectural departure from ELENA's design, in which acquisition, storage, and offline analysis are tightly coupled to a single MATLAB post-processing script; ParkinSense's runtime, motion pipeline, PPG pipeline, step pipeline, and dashboard are each separately importable modules, which is what allows the PPG and step-tracking pipelines to have been added on the `develop` branch without any change to the wire protocol or to the motion pipeline's own code.

B. Motion Pipeline

The motion pipeline operates on a 4-second rolling window of 104 Hz accelerometer and gyroscope samples, chosen to give stable spectral resolution while keeping detection latency low. This window length is shorter than the multi-minute batch recordings ELENA analyzes offline, and is a deliberate choice to support a live state estimate rather than a session-level average.

Gravity removal. A low-pass estimator subtracts the gravity component from the accelerometer channel, separating static wrist orientation from dynamic motion. ELENA's pipeline does not perform an equivalent gravity-separation step prior to its FFT, relying instead on its 1 Hz high-pass filter to suppress the slowest components of postural drift; ParkinSense separates orientation from dynamics explicitly, which keeps the downstream notch and band-pass stages operating on a cleaner dynamic-only signal.

Notch filtering. A digital 50 Hz notch filter is applied to both accelerometer and gyroscope channels to suppress mains interference, a stage with no counterpart in ELENA's filter chain.

Band-pass filtering. A Butterworth band-pass filter isolates the 4.0–6.5 Hz Parkinsonian resting-tremor band, attenuating slow postural drift below it and high-frequency vibration above it. This is a narrower and more targeted passband than ELENA's single low-pass-then-FFT approach, which retains a broader 1–12 Hz range and relies on peak selection after the fact rather than a hardware-matched passband before feature extraction.

Motion context classification. A `MotionContext` classifier labels each window `REST`, `LOW MOTION`, or `ACTIVE`, allowing the downstream detector to discount tremor-band energy that coincides with intentional movement. This classifier has no equivalent in ELENA, whose authors instead flag voluntary-motion artifact during the writing and waving tasks as an open, unresolved problem in their discussion. ParkinSense's motion-context label is precisely the missing piece that a per-axis FFT peak-pick cannot supply on its own, and it is reused, not recomputed, by the other two pipelines.

Feature extraction. Per gyroscope axis, RMS, dominant frequency, in-band energy ratio, spectral entropy, spectral centroid, zero-crossing rate, peak magnitude, and frequency stability are computed – eight features per axis, or twenty-four features per window in total, substantially more descriptive than the single per-axis dominant-frequency value ELENA extracts.

Tremor detection and confidence. All three gyroscope axes are scored jointly on in-band frequency, band-ratio

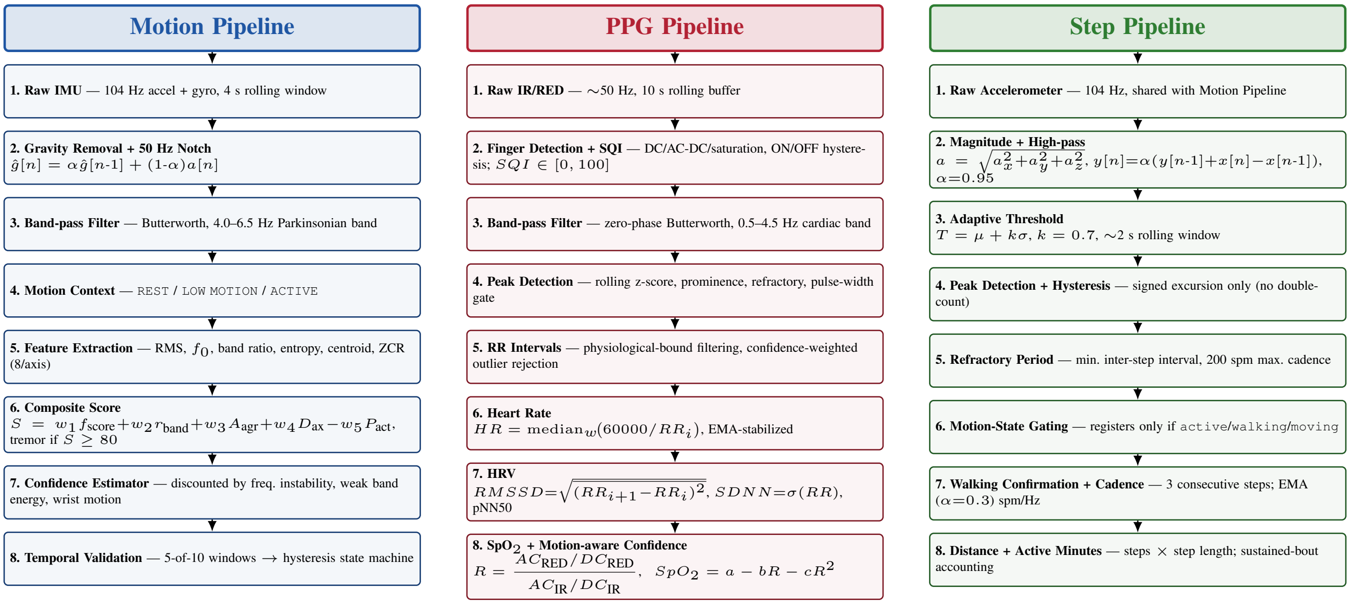


Fig. 2: Detailed stage-by-stage view of the three synchronized processing pipelines, annotated with the core algorithmic relations used at each stage. All three consume the same decoded BLE sample stream (Fig. 1); the Motion Pipeline's context classifier (stage M4) additionally gates Step Pipeline stage S6 and discounts PPG Pipeline stage P8.

strength, cross-axis agreement, axis dominance, and an active-state penalty; a composite score of 80/100 is required to flag a window as tremor-positive. A separate confidence estimator, discounted by frequency instability, weak band energy, or excessive wrist motion, is maintained independently of the binary tremor flag so that "is this tremor" and "how sure are we" remain decoupled – a distinction that a single dominant-frequency-per-axis output, as reported by ELENA, does not expose to a downstream consumer of the data.

Temporal validation and state machine. A rolling 10-window history requires five positive windows before tremor is confirmed, and a hysteresis-based state machine (`NO TREMOR` → `POSSIBLE` → `CONFIRMED` → `RECOVERY` → `NO TREMOR`) prevents rapid oscillation between states from transient movement. This closed-loop temporal filter is the mechanism that lets ParkinSense report a stable, continuously updated clinical state rather than a single offline frequency number per recording session.

The pipeline emits tremor score, dominant frequency, severity category, tremor burden, band ratio, axis agreement/dominance, and a rest index as its digital biomarkers, and additionally exposes its motion-context label to the step-tracking pipeline and its activity state to the PPG pipeline's confidence-fusion stage.

C. PPG Pipeline

The PPG pipeline maintains a rolling 10-second buffer per optical channel (IR and RED), sampled at the sensor's native rate (∼50 Hz). This entire pipeline is a capability that ELENA's single-IMU hardware cannot support at any level of algorithmic sophistication, since it has no optical sensor to source it from; we describe it in full because it is one of the two structural additions – alongside the step-tracking pipeline – that differentiate ParkinSense's contribution from an incremental improvement to ELENA's existing sensing modality.

Finger detection. Per-sample finger presence is assessed from DC level, AC/DC ratio, saturation, and cardiac-band content, and debounced with asymmetric ON/OFF hysteresis so that momentary contact loss does not flicker the reading.

Signal quality and sensor status. A Signal Quality Index (SQI, 0–100) is computed from spectral concentration, harmonic ratio, perfusion index, and pulse repeatability, and combined with DC level and clipping information to classify the channel as `NO_FINGER`, `LOW_SIGNAL`, `POOR_SIGNAL`, `GOOD`, `HIGH_SIGNAL`, or `SATURATED`.

Filtering and peak detection. A zero-phase Butterworth band-pass isolates the 0.5–4.5 Hz cardiac band. Systolic peaks are located via rolling z-score normalization, local energy gating, adaptive prominence scaling, a refractory period, and a pulse-width gate, and each accepted beat is scored on prominence, SNR, width, symmetry, slope, and RR consistency.

RR intervals, heart rate, and stabilization. Beat-to-beat intervals are filtered against physiological bounds and cleaned with confidence-weighted outlier rejection. A confidence-weighted median of the cleaned RR intervals yields per-window heart rate, which is then smoothed with an exponential moving average; low-confidence windows are withheld rather than allowed to perturb the published value.

Heart-rate variability. Once sufficient clean RR intervals are available, RMSSD, SDNN, mean RR, and pNN50 are computed following standard time-domain HRV definitions [10], each with its own confidence and artifact-percentage diagnostic.

SpO_2 estimation. A beat-by-beat ratio-of-ratios estimate, with RED/IR peak-alignment validation, a calibration mapping, and a `SEARCHING` → `LOCKED` → `TRACKING` → `LOST` signal-lock state machine, produces a rate-limited, confidence-weighted SpO_2 estimate. This value is explicitly treated as research-grade rather than clinically validated pulse oximetry.

Motion-aware confidence fusion. Overall heart-rate confidence is discounted according to the motion pipeline's reported activity state, so that a walking or running window is trusted less than a resting one without being discarded outright – the key point of cross-pipeline coupling on the cardiac side, and a direct, concrete instance of the shared-motion-state design principle argued for in Section II-E.

D. Step-Tracking Pipeline

The step-tracking pipeline runs off the same 104 Hz accelerometer stream consumed by the motion pipeline, requiring no additional acquisition hardware and, again, no counterpart in ELENA's single-purpose tremor architecture.

Magnitude and high-pass filtering. The three accelerometer axes are combined into a single magnitude, $a = \sqrt{a_x^2 + a_y^2 + a_z^2}$, and a first-order high-pass filter ($\alpha = 0.95$) removes gravity and slow postural drift, leaving the oscillatory component associated with footstrike impacts.

Adaptive threshold. A rolling $\sim$2-second buffer feeds a self-tuning threshold $\text{mean}+k\sigma$ ($k = 0.7$), clamped between a floor and ceiling, allowing the same detector to operate across a range of gait intensities without per-user hand tuning.

Peak detection and hysteresis. Only the signed, positive-going excursion triggers a step count, since using the rectified magnitude would double-count each oscillation; the detector re-arms only after the signal falls below a hysteresis band, preventing a single step from being counted twice as it decays.

Refractory period. A minimum inter-step interval, derived from a configurable maximum cadence (200 steps/min by default), rejects candidate steps arriving faster than is physiologically plausible.

Motion-state gating. Steps are registered only while the motion pipeline's context classifier reports `active`/`walking`/`moving`. This is the mechanism that prevents hand tremor or incidental device handling from being miscounted as steps – the step counter deliberately does not re-solve "is the body moving," relying instead on the motion pipeline's existing classification. We consider this gating mechanism the single clearest concrete illustration of why the three-pipeline architecture is more than three separate features bundled into one firmware image: it is precisely the failure mode ELENA's authors identify as an open problem for their own writing/waving tasks, addressed here for a different but related channel.

Walking confirmation, cadence, and distance. A walking flag is set only after three consecutive steps land at a gait-consistent interval, and clears after a walking timeout once steps stop arriving. Instantaneous step intervals are smoothed with an EMA ($\alpha = 0.3$) to yield cadence in steps/min and Hz. Distance accumulates as step count times a configurable step length, left as a constructor parameter for future per-user calibration (e.g., from height).

TABLE II: ParkinSense Hardware and Software Specification

Category	**Component**	**Key Specification**
MCU	Seeed XIAO nRF52840 Sense	ARM Cortex-M4F, BLE 5.0
IMU	SparkFun LSM6DS3	6-axis, 104 Hz ODR
Optical sensor	MAX30102	IR + RED, $\sim$50 Hz
Battery	3.7 V 700 mAh Li-ion (902035)	USB-C charging
BLE payload	Binary packet	10 samples/packet, $\sim$104 Hz stream
Motion window	4 s rolling	FFT / feature resolution
PPG window	10 s rolling	HR/HRV/SpO_2 estimation
Step window	$\sim$2 s rolling	Adaptive threshold
Runtime	Python 3.11, `numpy`/`scipy`/`bleak`	Continuous, <100 ms latency after window close
Dashboard	Plotly Dash	100 ms refresh, file-based decoupling

Active-minute accounting. A stricter criterion than walking: a bout accrues active minutes only once it has been sustained past a minimum duration and cadence threshold, preventing brief movements from inflating activity totals.

Confidence heuristics. Step confidence is derived from how far the triggering peak cleared the adaptive threshold; walking confidence is derived from the coefficient of variation of recent step intervals, capturing gait regularity.

E. Digital Biomarker Fusion and Logging

All three pipelines publish to a shared logging layer: every processed sample is appended to a CSV file (raw IMU/PPG channels, tremor metrics, cardiac metrics, and activity metrics together), and a JSON metrics file is updated for dashboard consumption. The dashboard itself is intentionally decoupled from the pipelines – it reads only the exported files and never imports pipeline code directly – so that the sensing and analytics layers can be tested, replayed, or replaced independently of the presentation layer. Where ELENA stores raw acceleration locally on a microSD card and separately uploads it to a cloud MATLAB pipeline for post-hoc analysis, ParkinSense's CSV/JSON export already contains fully processed, window-level biomarkers from all three pipelines at acquisition time, making the exported dataset immediately usable for a future supervised machine-learning pipeline without a separate offline reprocessing step.

IV. Hardware and Software Specification

Table II summarizes the ParkinSense hardware and software stack, and Table III contrasts it directly against ELENA's published hardware bill of materials.

The choice of a BLE-based local link rather than ELENA's Wi-Fi/cloud path is deliberate. A Wi-Fi/cloud path introduces network dependency, upload latency, and a requirement for internet connectivity that a purely local BLE stream to a nearby host does not; ELENA's own limitations section acknowledges this dependency and proposes a future hybrid

TABLE III: Hardware Comparison: ParkinSense vs. ELENA

Aspect	ELENA [28]	ParkinSense
Inertial sensor	MPU6050 (accel + gyro)	LSM6DS3 (accel + gyro), 104 Hz ODR
Optical sensor	None	MAX30102 (IR + RED)
MCU	ESP32 (Adafruit HUZZAH32)	nRF52840 (nRF52840 Sense)
Radio link	Wi-Fi to cloud server	BLE 5.0, 104 Hz local stream
On-board analysis	None (offline MATLAB)	Real-time, on-device (motion, PPG, step)
Local storage	MicroSD	CSV/JSON on host runtime
Feedback	OLED + buzzer	Live Plotly Dash dashboard
Latency to result	Post-session (offline)	<100 ms after 4 s window

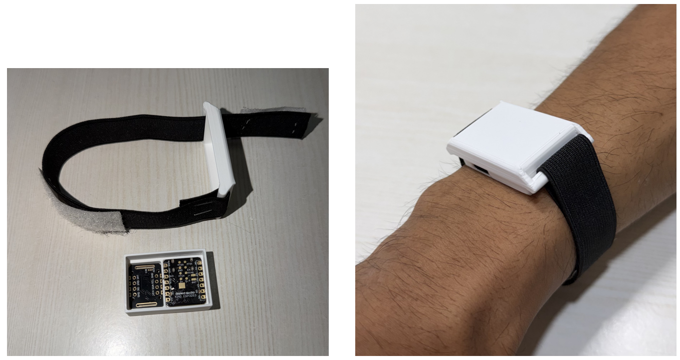
edge/cloud architecture as a fix, which is effectively the architecture ParkinSense already implements. The choice of the LSM6DS3 over the MPU6050 is driven by its higher-stability output data rate and lower noise density at the 104 Hz sampling rate ParkinSense's motion pipeline requires for its 4-second windowing, and the addition of the MAX30102 is the physical prerequisite for the entire PPG pipeline described in Section III-C, which has no equivalent in ELENA's bill of materials at all.

A. Wearable Prototype Implementation

The signal-processing pipelines described in Section III are not evaluated on a bare sensor breakout but on a complete, self-contained wearable built for continuous daily use. All electronics – the Seeed XIAO nRF52840 Sense, the SparkFun LSM6DS3 IMU, and the MAX30102 optical sensor – are integrated inside a custom-designed, 3D-printed wrist housing sized to sit flush against the wrist rather than around it, so that the finished unit reads as a purpose-built wearable device rather than an exposed development board strapped to the arm. The MAX30102 is seated in the underside of the housing directly against the skin to keep optical coupling stable as the arm moves. Power comes from a rechargeable 3.7 V, 700 mAh Li-ion cell charged over USB Type-C through the charge-management circuitry already built into the XIAO nRF52840 Sense, so the wearer can top up the device without opening the enclosure or handling the cell directly. Between the enclosed form factor and USB-C charging, the prototype is intended to be worn through ordinary daily activity rather than only during short, supervised bench sessions, which is a prerequisite for the free-living data collection proposed as future work in Section VII. Figure 3 illustrates the fabricated wearable prototype both as a standalone device and during wrist wear.

B. Real-Time Visualization Dashboard

The Plotly Dash layer introduced in Section III-E is delivered as two views over the same CSV/JSON export rather than two separate applications. An engineering view surfaces live BLE connection diagnostics, raw and filtered motion and PPG waveforms, and per-pipeline signal-quality indicators for development and bench validation. A clinical view instead reorganizes the identical underlying biomarkers into three groupings – neurological (tremor state and related motion metrics), activity (cadence, steps, distance), and vitals (heart rate, SpO_2, HRV) – intended for someone reviewing the data rather than debugging the firmware. Because both views read from the same exported files at the 100 ms refresh interval given in Table II, the motion, PPG, and activity series stay synchronized on screen exactly as they were synchronized at capture time. The dashboard can also replay a previously logged CSV session through the same rendering path used for live data, so a recorded session can be stepped through after the fact without re-running the sensing hardware. Figure 4 illustrates the dashboard during both normal operation and tremor detection, highlighting synchronized visualization of neurological, cardiac, and activity biomarkers.



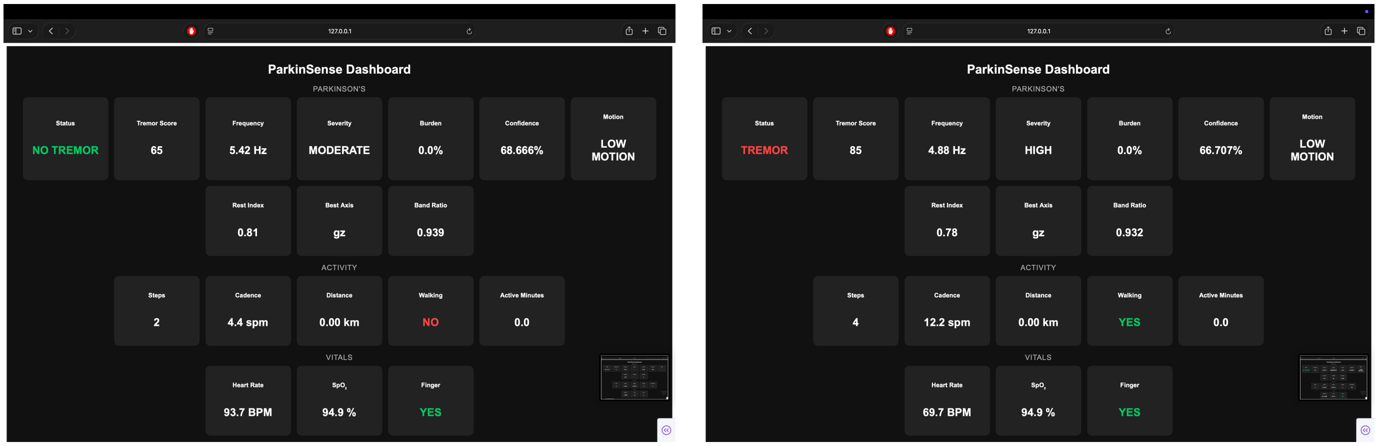
(a) Prototype enclosure (b) Prototype worn on the wrist

Fig. 3: ParkinSense wearable prototype showing the custom 3D-printed enclosure, the housed XIAO nRF52840 Sense, LSM6DS3 IMU, MAX30102 optical sensor, and USB Type-C charging port, and the complete device during wear.

V. Experimental Evaluation

A. Experimental Setup

ParkinSense was evaluated as a complete wearable prototype under controlled laboratory conditions rather than on a bare sensor breakout. The assembled device, consisting of a Seeed XIAO nRF52840 Sense, SparkFun LSM6DS3 IMU, and MAX30102 optical sensor housed in the enclosure described in Section IV-A, streamed data continuously over BLE to the Python runtime described in Section III. All three signal-processing pipelines operated in real time, with no offline post-processing step. The evaluation reported in this section targets engineering performance – communication reliability, pipeline functional correctness, dashboard synchronization, and power consumption – under representative motion conditions (rest, simulated tremor, walking, and incidental arm movement). Clinical validation against PD patients and reference medical instrumentation has not yet been performed and is treated separately, as future work, in Section VII.



(a) Normal activity (no tremor) (b) Detected Parkinsonian tremor

Fig. 4: ParkinSense clinical dashboard during live monitoring. The dashboard displays synchronized neurological, cardiac, and activity biomarkers under normal conditions and during tremor detection.

TABLE IV: Measured BLE Communication Performance

Metric	Measured Value
Sampling rate	104 Hz
Samples per BLE packet	10
Packet size	208 Bytes
BLE version	BLE 5.0
Dashboard refresh	100 ms
End-to-end processing latency	<100 ms after window close
Packet loss	None observed in continuous testing

B. BLE Communication Performance

Table IV summarizes the measured communication characteristics of the BLE link described in Section IV. Across continuous multi-hour bench sessions, the connection remained stable, the runtime's packet decoder produced no decode failures, and no packet loss was observed. The accelerometer, gyroscope, and PPG channels arrived pre-synchronized within each notification (Fig. 5), so no additional cross-channel time-alignment step was required downstream.

C. Motion Pipeline Evaluation

The motion pipeline (Section III-B) was exercised under four representative bench conditions: the device stationary on a table, a motorized fixture producing a simulated 5 Hz oscillation within the Parkinsonian band, unconstrained walking, and unconstrained random arm movement. Table V summarizes the expected and observed classifier behavior. In all four conditions the motion-context classifier and tremor state machine produced the expected label, correctly distinguishing the simulated resting-tremor frequency from both walking and non-rhythmic voluntary motion, and the hysteresis logic prevented single noisy windows from flipping the reported state.

TABLE V: Motion Pipeline Functional Test Results

Test Condition	Expected	Observed
Device stationary	No tremor	Correct
Simulated 5 Hz oscillation	Tremor detected	Correct
Walking	`ACTIVE`/walking state	Correct
Random arm movement	`ACTIVE` state	Correct

D. PPG Pipeline Evaluation

The PPG pipeline (Section III-C) was assessed for functional correctness during bench sessions with the sensor worn against the wrist. Table VI summarizes the qualitative observations. Heart rate tracked the expected pulse rate stably once the signal-lock state machine reached `LOCKED`/`TRACKING`; finger/contact detection reliably distinguished worn from not-worn states; the SQI classifier correctly separated clean signal windows from motion-corrupted ones; and, consistent with the motion-aware confidence-fusion design of Section III-C, reported heart-rate confidence dropped during walking and random-motion trials relative to seated-rest trials, without the value being discarded outright. The SpO_2 estimate remained stable during resting, well-perfused conditions, consistent with its intended role as a research-grade rather than clinically calibrated estimate.

E. Step-Tracking Evaluation

The step-tracking pipeline (Section III-D) was tested against standing, walking, simulated hand tremor, and deliberate hand-shaking conditions to specifically probe the motion-state gating mechanism described in Section III-D. Table VII summarizes the results. Standing produced no counted steps; walking produced a correctly incrementing step count with plausible cadence; and, critically, both the tremor-simulator condition and deliberate hand-shaking produced no counted

TABLE VI: PPG Pipeline Functional Observations

Parameter	Observation
Heart rate	Stable, tracks pulse after signal lock
Finger detection	Reliable worn/not-worn discrimination
Signal Quality Index	Correctly classifies clean vs. motion-corrupted windows
Motion-aware confidence	Reduced during walking/random motion
SpO_2	Stable research-grade estimate at rest

TABLE VII: Step-Tracking Functional Test Results

Activity	Expected	Result
Standing	0 steps	Correct
Walking	Steps counted	Correct
Simulated tremor	No steps	Correct
Hand shaking	No steps	Correct

steps, since the motion-context classifier did not report `active`/`walking`/`moving` during either. This is one of the platform's more direct pieces of evidence that cross-pipeline gating functions as intended: a step counter without this gate would be expected to register spurious steps from wrist oscillation alone, which is exactly the class of artifact ELENA's own authors report as an open problem for their comparable active-task recordings (Section II-E).

F. Dashboard Evaluation

Both dashboard views described in Section IV-B – the engineering view and the clinical view – were exercised during live sessions and in replay mode over previously logged CSV files (Fig. 4). In both modes, the motion, PPG, and activity series remained synchronized on screen at the 100 ms refresh rate reported in Table IV, and replay of a logged session reproduced the same rendering path used for live data without requiring the sensing hardware to be reconnected.

G. Power Evaluation

Table VIII reports the measured power characteristics of the assembled prototype, obtained with a Nordic Power Profiler Kit II rather than assumed from component datasheets. Active current during continuous sensing and BLE streaming averaged approximately 7 mA, which against the 700 mAh cell described in Section IV-A corresponds to roughly 100 hours of continuous runtime between charges. The firmware drops into a deep-sleep state, with current draw in the microampere range, whenever the MAX30102 stops detecting skin contact, reusing the same contact-detection logic already described for the PPG pipeline, and resumes full-rate sensing and streaming automatically as soon as skin contact is restored.

H. Discussion

Unlike a closed-loop control system, whose performance can be verified against a physical reference within a single bench session, a longitudinal biomarker platform's value depends on agreement with clinical ground truth collected over

TABLE VIII: Measured Power Characteristics of the ParkinSense Wearable Prototype

Parameter	Measured Value
Active current (streaming)	≈7 mA (Nordic Power Profiler Kit II)
Battery	3.7 V, 700 mAh Li-ion, USB-C rechargeable
Continuous runtime	≈100 hours
Sleep trigger	Loss of skin contact (MAX30102)
Sleep current	Microampere range
Wake trigger	Restoration of skin contact (automatic)

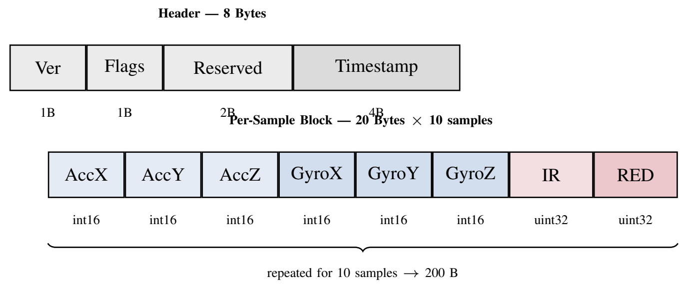


Fig. 5: ParkinSense BLE notification layout: an 8-byte header followed by ten 20-byte inertial+optical sample blocks.

time, which is why we report ParkinSense's status honestly rather than substitute bench results for a clinical claim, and why we do not yet claim parity with, let alone superiority over, ELENA's completed nine-patient clinical pilot on any patient-level outcome measure. The present evaluation demonstrates engineering feasibility rather than clinical performance: the communication subsystem maintained continuous streaming at 104 Hz with no observed packet loss, all three processing pipelines produced the expected qualitative behavior under controlled bench conditions, the motion-gating mechanism correctly suppressed tremor- and handling-induced false steps, and the wearable achieved approximately 100 hours of continuous operation on a single charge. Nevertheless, the algorithms have not yet been validated against clinician-annotated Parkinson's disease data or medical-grade reference instrumentation (a reference pulse oximeter, a chest-strap ECG, or a reference pedometer). Consequently, every metric the platform emits is explicitly labeled research-grade in the documentation and dashboard, and the platform is not positioned as a diagnostic device. Closing this specific gap – bench-level frequency-accuracy validation directly comparable to ELENA's published protocol, followed by an IRB-approved patient pilot – is the subject of the future-work plan in Section VII.

VI. Comparative Analysis

ParkinSense, ELENA, PERFORM, and TremoSense address adjacent but distinct problems within PD wearable technology and are not directly comparable on a single performance axis: TremoSense is a closed-loop actuation system evaluated by tremor rejection percentage on a bench simulator;

TABLE IX: Structural Comparison: ParkinSense vs. Related Systems

System	**Modalities**	**Analysis**	**Validation**
Single-modality tremor wearables [6], [7]	Motion only	Varies by study	Clinical (varies)
Consumer PPG wearables	Cardiac only	On-device	Consumer-grade
ELENA [28]	Motion only (accel.+gyro)	Offline, MATLAB FFT	9 PD + 9 control clinical pilot; <0.5% freq. deviation vs. Apple Watch
PERFORM [17]	Multi-sensor	Clinical decision support system	System-level clinical framework
TremoSense / [20]	Motion + control	Real-time, on-device	Bench simulator, % rejection
ParkinSense	Motion + PPG + activity	Real-time, on-device, joint multi-axis	Engineering/bench evaluation complete; clinical pilot planned

TABLE X: Algorithmic Comparison of the Motion/Tremor Channel: ParkinSense vs. ELENA

Stage	**ELENA [28]**	**ParkinSense**
Pre-filtering	Butterworth low-pass + 1 Hz high-pass	Gravity removal + 50 Hz notch + 4.0–6.5 Hz band-pass
Frequency analysis	Per-axis FFT, offline	Per-axis dominant frequency, spectral entropy, centroid, ZCR, on-device
Axis handling	Independent; highest peak selected	Jointly scored; cross-axis agreement + dominance terms
Decision	Peak read from spectrum	Composite 0–100 score against an 80-point threshold
Confidence	Not separately reported	Independent confidence estimator (frequency stability, band energy, motion)
Temporal logic	None (single-window batch)	10-window history, 5-of-10 rule, hysteresis state machine
Context awareness	None	`REST`/`LOW MOTION`/`ACTIVE` classifier feeding an active-state penalty
Latency	Post-session (minutes to hours)	<100 ms after each 4 s window

PERFORM is a multi-sensor clinical monitoring framework evaluated at a systems level rather than as a single reproducible low-cost wearable; ELENA is an offline-analyzed, single-modality tremor logger evaluated by device-level frequency accuracy and a nine-patient clinical pilot; and ParkinSense is a real-time, on-device, multi-modal monitoring platform currently evaluated for engineering feasibility and internal signal-processing correctness (Section V). Table IX summarizes the structural differences, and Table X summarizes the specific algorithmic differences between ParkinSense's motion pipeline and ELENA's frequency-analysis pipeline, since the motion channel is the one directly comparable capability between the two systems.

The structural contribution of ParkinSense relative to single-modality tremor wearables such as ELENA is therefore twofold. On the hardware side, the addition of a synchronized optical channel and a low-latency local radio link makes two further biomarker domains (cardiac and activity) available from the same wrist-worn package without any additional acquisition hardware for the step-tracking pipeline. On the algorithm side, the explicit cross-pipeline coupling means the step-tracking pipeline consumes the motion pipeline's context classification directly, and the PPG pipeline's confidence output is discounted by the same classification, so that a single motion-state estimate is derived once, validated once, and reused three times, rather than each channel independently re-deriving (or, as in ELENA's case for its writing/waving tasks, failing to derive) its own notion of voluntary motion. ELENA's contribution is complementary and, at this stage, more clinically mature: it demonstrates that a low-cost IMU-based wearable can match a commercial reference device on frequency accuracy and can be deployed in an IRB-approved patient study, which is precisely the kind of evaluation ParkinSense has not yet undergone. A natural and concrete extension of this comparison, which we intend to carry out as the first step of the future-work plan in Section VII, is to benchmark ParkinSense's motion pipeline against ELENA's own validated Apple Watch/vibration-table protocol directly, since both systems target the same 3–8 Hz resting-tremor band and both were first validated using a mechanical vibration source rather than free-living data.

VII. Conclusion and Future Work

This paper presented ParkinSense, a three-pipeline wearable platform that jointly derives motion, cardiac, and activity digital biomarkers from a single synchronized inertial-plus-optical sensor stream, with the activity pipeline explicitly gated by the motion pipeline's context classifier to prevent tremor and handling artifacts from being miscounted as gait. We argued in detail, against the specific and rigorously validated benchmark set by the ELENA project, that ParkinSense's improvements are structural rather than incremental: an added optical sensing modality and a low-latency local radio link on the hardware side, and joint multi-axis scoring, an independent confidence estimator, a temporal hysteresis state machine, and cross-pipeline motion-state gating on the algorithmic side. We reported the platform's experimental evaluation without overstating it: BLE communication, all three pipelines, the dual dashboard views, and the power/battery behavior were validated on the assembled wearable prototype under controlled bench conditions, but clinical-grade validation against labeled ground truth – including the kind of nine-patient pilot ELENA has already completed – has not yet been performed, and every emitted metric is treated as research-grade pending that validation.

This validation status rests on a device that already exists as a wearable product, not only as a bench setup. Housed in a custom 3D-printed enclosure and powered by a rechargeable 700 mAh Li-ion cell charged over USB Type-C through the XIAO nRF52840 Sense, the prototype sustains roughly 100 hours of continuous operation on a single charge, draws approximately 7 mA while actively streaming, and falls back to a microampere-range sleep state – with automatic wake-up – whenever skin contact is lost. Paired with the dual engineering/clinical dashboard views described in Section IV-B, this combination of a finished form factor and multi-day battery life is what makes the free-living data collection proposed below a practical next step rather than only a bench-testing exercise.

Future work is structured explicitly to close the specific gap identified in Section II-E and reaffirmed by the honest limitations of Section V. First, a bench-level frequency-accuracy validation directly comparable to ELENA's published protocol: a motorized tremor simulator and an MTS-style vibration table, following a calibration procedure modeled on ELENA's own six-position and dynamic vibration-table protocol, benchmarked against the same Apple Watch reference ELENA used, so that ParkinSense's device-level frequency accuracy can be reported as a directly comparable number rather than an incommensurable one. Second, a small IRB-approved clinical pilot with PD patients and age-matched controls, collecting concurrent clinician tremor ratings and reference physiological measurements – a reference pulse oximeter for SpO_2, a chest-strap ECG for HRV ground truth, and a reference pedometer for step-count accuracy – against which the CSV/JSON logs can be scored, with the false-step rate during confirmed tremor episodes as a specific outcome measure not present in ELENA's own evaluation, since it directly tests whether the motion-state gating mechanism achieves its stated purpose. Third, release of the resulting labeled dataset alongside the existing replay framework, enabling machine-learning roadmap items already scoped in the project: personalized adaptive thresholds, activity-type classification, and exploratory bradykinesia/dyskinesia detection. On the hardware side, planned additions include battery-level monitoring, further power optimization for longer continuous wear, and a mobile companion application.

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